

Data Pipelines & Kafka: Review Questions

Powerpoints - Modules 1-9

1. Fill in the blank: Ward Cunningham's metaphor compares shipping a quick version of code to taking out a loan. The ongoing cost of that shortcut is similar to the _____ accrued for a loan.

Answer: interest

2. True/False: Technical debt refers to code that is broken and doesn't work.

Answer: False. Debt is working code that is expensive or risky to change.

3. Fill in the blank: One of the main differences between a program bug and a technical debt bug is _____ type bugs usually do not involve the program failing.

Answer: Program Bugs; Tech Debt Bugs

4. True/False: Technical debt is only accrued during the initial implementation of a project.

Answer: False

5. Fill in the blank: Batch processing works on a _____ dataset, while real-time processing works on an _____ stream.

Answer: bounded; unbounded

6. True/False: Batch failures are usually silent, while real-time stream failures are usually visible.

Answer: False. It's the reverse: batch failures are usually visible and stream failures can be silent.

7. True/False: "Batch vs real-time" and "event-driven vs request-driven" architecture questions are asking the same thing just asked in two different ways.

Answer: False. They are all independent in terms of consideration.

8. Fill in the blank: In the request-driven example, OrderService makes three blocking calls of about 400ms each, so the order takes roughly _____ seconds in total.

Answer: 1.2 (about 1200ms)

9. True/False: In order to use an event based architecture you need a full blown product such as Kafka.

Answer: False. This architecture can be supported by an in-memory, single-process, and inside a JVM.

10. Fill in the blank: A Kafka topic is split into a fixed number of independent, ordered, append-only logs called _____.

Answer: partitions

11. True/False:: The partition size is fixed and cannot be adjusted.

Answer: False

12. True/False: Kafka guarantees a single, total order of events across the entire topic regardless of key

Answer: False. Order is guaranteed only per key, within a partition.

13. True/False: An offset is used to indicate the position of a record in a partition.

Answer: True

14. True/False: A replication factor of 1 is fine for a single-broker dev setup but never acceptable in production.

Answer: True. In production want 3 or more in case the broker fails and you still want data available.

15. Fill in the blank: It is often a practice that when a producer issues a message it waits for _____ to ensure the message has been received.

Answer: Confirmation

16. True/False: In the real-broker demo, the producer code decided each event's partition with its own floorMod logic.

Answer: False. The broker assigned partitions and offsets. floorMod was the in-memory simulator.

17. True/False: You define an initial number of partitions for your Kafka deployment. This count can be safely decreased later, but it cannot be increased.

Answer: False. It's the reverse: it can be increased but not safely decreased, and increasing it changes which partition a key hashes to.

18. Fill in the blank: A `trade-events` topic feeds four systems: Settlement, the Risk dashboard, and Fraud detection. Each need events ordered per account, and the Compliance log needs no ordering at all. The best partition key is _____.

Answer: `accountId`

19. Fill in the blank: What type of architecture would best fit a scenario where once a trade is placed, the trade amount needs to be communicated to 5 other systems.

Answer: Event-Based/Kafka

20. Fill in the blank: Name two of the three questions that help to determine the data quality of every field that is processed whether batch or event-based.

Answer: Completeness/Validity/Plausibility

21. True/False: It is good practice to substitute exception handling for validation.

Answer: False. Good validation results in prudent exception handling.

22. Fill in the blank: An operation that produces the same result no matter how many times it runs is called _____.

Answer: idempotent

23. Fill in the blank: The three data quality dimensions covered are completeness, validity, and _____.

Answer: plausibility

24. True/False: When processing a batch record, it is good practice to list the number of the row that failed when doing an error log report.

Answer: True

25. True/False: In most cases in a batch scenario, when records are not processed, they should just wait until next day to attempting to process them again.

Answer: False

26. True/False: When GenAI explains a Kafka error, its answer should be accepted as a verdict rather than checked against the code.

Answer: False. Should always use a second source and team feedback to review GenAI explanations.

27. True/False: A try/catch that swallows bad rows destroys the evidence a problem ever existed.

Answer: True

28. True/False: The publishing of a trade event and the processing of the trade event by a subscriber should be done in the same application.

Answer: False

29. Fill in the blank: If your broker has a `SerializationException` occur such as 'Size of data received by `IntegerDeserializer` is not 4', this is an example of a _____ type of error.

Answer: `RecordDeserializationException`

30. Fill in the blank: If your broker has a `SerializationException` occur such as 'Can't convert value of class `Order` to class `String`. This is an example of a _____ type of error.

Answer: `StringSerializationException`

31. Fill in the blank: What is the name of the coordination service that is used by the older implementation of Kafka to track broker metadata and runs as a separate process.

Answer: `ZooKeeper`

32. Fill in the blank: The name of the current coordination service that replaced `ZooKeeper` with no second docker container required is _____.

Answer: `Kafka Raft` or `KRaft`

33. True/False: Kafka ships its admin tooling as shell scripts.

Answer: `True`

34. True/False: Kafka ships a built-in Java API in the JDK.

Answer: False

35. Fill in the blank: Apache Kafka's official client library which can be added as a maven dependency is named _____.

Answer: Kafka-Clients

36. Fill in the blank: The three main parts of a Kafka _____.

Answer: Kafka-Clients

37. True/False: The offset for a Kafka partition acts as an indicator of where the message being processed was located in the partition

Answer: True

38. Fill in the blank: In Kafka a _____ is defined by the consumer configuration and identifies which consumer group that consumer belongs to.

Answer: [group.id](#)

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40. True/False: In Kafka if you change the [group.id](#) for a set of consumers and then rerun the consumers, it will only process new messages that have been come into the queue.

Answer: False. It creates a new consumer group that has no previously committed offsets for the topic.